**Full Name:** _____

**Variant 1**

**Duration:** 120 Minutes

Innopolis University
Introduction to Optimization
Fall 2024
Midterm-Exam
26.10.2024

You need to perform this exam alone and without the use of any equipment or book, apart from this booklet, **A5** cheat sheet, a calculator and pens. Please do not consult any person or use any equipment, otherwise you will be disqualified from this exam at the very first attempt.
Good Luck!

Grade Table (for teacher use only)

| Question: | 1 | 2 | 3 | 4 | 5 | 6 | Total |
|---|---|---|---|---|---|---|---|
| Points: | 2 | 6 | 5 | 4 | 4 | 4 | 25 |
| Score: | | | | | | | |

1. (2 points) Write the dual problem for the following primal problem:

   Maximize

   $$z = x_1 + 5x_2 + 3x_3$$

   subject to

   $$\begin{aligned} x_1 + 2x_2 + x_3 &\leq 3 \\ 2x_1 - x_2 &\leq 4 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

2. Consider the transportation problem having the following parameter table:

| | | Destination | | | |
|---|---|---|---|---|---|
| | | 1 | 2 | 3 | Supply |
| Source | 1 | 6 | 3 | 5 | 4 |
| | 2 | 4 | M | 7 | 3 |
| | 3 | 3 | 4 | 3 | 2 |
| Demand | | 4 | 3 | 2 | |

(a) (2 points) Use Vogel's approximation method manually to select the first basic variable for an initial BF solution.

(b) (2 points) Use Russell's approximation method manually to select the first basic variable for an initial BF solution.

(c) (2 points) Use the northwest corner rule manually to construct a complete initial BF solution.

3. Consider the following LP.

   Maximize

$$F(x_1, x_2) = 2x_1 + 2x_2$$

   subject to

$$\begin{cases} 3x_1 - 2x_2 \leq 6, & (1) \\ 3x_1 + 4x_2 \leq 24, & (2) \\ x_1 \geq 0, & (3) \\ x_2 \geq 0 & (4). \end{cases}$$

   (a) (1 point) Find the optimal solution using the graphical method.

   (b) (2 points) Find the optimal solution using Simplex method.

   (c) (1 point) Determine the optimality condition for $c_1$ and $c_2$ that will keep the optimum unchanged.

   (d) (1 point) Determine the optimality ranges for $c_1$, assuming that the other coefficient is kept constant at its present value.

4. Use an interior-point algorithm find $\mathbf{X}^1 = (x_1^1, x_2^1)$ for the following LP

   Maximize

   $$F(x_1, x_2) = 2x_1 + 2x_2$$

   subject to

   $$\begin{cases} 3x_1 - 2x_2 \leq 6, & (1) \\ 3x_1 + 4x_2 \leq 24, & (2) \\ x_1 \geq 0, & (3) \\ x_2 \geq 0 & (4). \end{cases}$$

   Put that $\alpha = 0.5$.

   (a) (1 point) Express the problem in equation form and choose $\mathbf{X}^0$ that lies in the interior of the feasible region.

   (b) (2 points) Substitute the corresponding matrices into the formula of $\mathbf{P}$. Skip calculations.

   (c) (1 point) Write the formula for $\mathbf{X}^1$.

5. (4 points) Table shows the last iteration of phase I of the following LP:

   minimize

$$z = -2x_1 + 5x_2 - 3x_3$$

   subject to

$$\begin{aligned} 2x_1 - 4x_2 + 4x_3 - s_1 + R_1 &= 28 \\ -3x_1 + 6x_2 - 3x_3 - s_2 + R_2 &= 24 \\ -3x_1 + 5x_2 + 2x_3 + s_3 &= 20 \\ x_1, x_2, x_3, s_1, s_2, s_3, R_1, R_2 &\geq 0. \end{aligned}$$

| Basic | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $R_1$ | $R_2$ | Sol. |
|---|---|---|---|---|---|---|---|---|---|
| $z$ | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 |
| $x_3$ | 0 | 0 | 1 | -1/2 | -1/3 | 0 | 1/3 | 1/2 | 22 |
| $x_1$ | 1 | 0 | 0 | -9/2 | -14/3 | -2 | 14/3 | 9/2 | 198 |
| $x_2$ | 0 | 1 | 0 | -5/2 | -8/3 | -1 | 8/3 | 5/2 | 114 |

   Determine optimal solutions $(x_1, x_2, x_3)$ by completing phase II.

6. Consider the following LP.

   Maximize

$$z = 9x_1 + 10x_2 + 16x_3$$

   subject to

$$\begin{aligned} 18x_1 + 15x_2 + 12x_3 &\leq 360, \\ 6x_1 + 4x_2 + 8x_3 &\leq 192, \\ 5x_1 + 3x_2 + 3x_3 &\leq 180, \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

   (a) (1 point) Express the problem in equation form.

   (b) (1 point) Express the problem in matrix form ($z = CX$ subject to $AX = b$,$X \geq 0$).

   (c) (1 point) Determine $X_{B_0}$, $C_{B_0}$.

   (d) (1 point) Which variable is entering and which variable is leaving.